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Ex 1: $X \sim \text{Bin}(n, \theta)$, $\theta \in (0, 1)$ We want to est. $\psi(\theta) = \frac{1-\theta}{\theta} = \frac{1}{\theta} - 1$
 $\downarrow$
 parametricQ) Do we have an unbiased est. of $\psi(\theta)$?(Note: Estimating $\psi(\theta)$ is equiv. to estimating $\frac{1}{\theta}$)A) If possible, let $T(X)$ be the unbiased est. of $\frac{1}{\theta}$

$$\Leftrightarrow E_{\theta}(T(X)) = \frac{1}{\theta} \quad \forall \quad 0 < \theta < 1 \quad (\mathbb{H}) = (0, 1)$$

$$\Leftrightarrow \sum_{i=0}^n T(i) \binom{n}{i} \theta^i (1-\theta)^{n-i} = \frac{1}{\theta} \quad \forall \theta \in (0, 1)$$

$$\Leftrightarrow \sum_{i=0}^n T(i) \binom{n}{i} \theta^{i+1} (1-\theta)^{n-i+1} = 1 \quad \forall \theta \in (0, 1)$$

 $\hookrightarrow$ quite literally impossible. (#) $\therefore \psi(\theta)$ doesn't have an unbiased est.# Show that $\psi(\theta)$ has unb. est. iff $\psi(\theta)$ is a poly. of deg. $\leq n$.Ex 2: $X \sim \text{Poi}(\theta)$, $\theta > 0$ Q) For what kind of $\psi(\theta)$, we have unb. est.?A) Let $T(X)$ be an unb. est. of $\psi(\theta)$.

$$\Leftrightarrow E_{\theta}(T(X)) = \psi(\theta) \quad \forall \theta > 0$$

$$\Leftrightarrow \sum_{i=0}^{\infty} T(i) \cdot e^{-\theta} \cdot \frac{\theta^i}{i!} = \psi(\theta) \quad \forall \theta > 0$$

$$\Leftrightarrow \sum_{i=0}^{\infty} \frac{T(i) \cdot \theta^i}{i!} = \psi(\theta) \cdot e^{\theta} \quad \forall \theta > 0$$

We know, e^{θ} is analytic (it has power series expansion)

$\therefore$ Check that $\psi(\theta)$ is (unbiasedly) estimable iff $\psi(\theta)$ is an analytic function.

- Note that, in both the examples our arguments lead to a way of constructing ~~an~~ unb. est. $T(X)$ for $\psi(\theta)$ when it exists.
- Also, $\therefore$ the power series coefficients of $\psi(\theta)$ are determined uniquely, in both the examples, the unbiased estimator will be unique and $\therefore$ it'll trivially be the UMVUE.

!Note! We have been implicitly assuming $E(X) < \infty$ for now.

Ex3: $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} f(x|\theta) = \frac{1}{\theta} e^{-\frac{x}{\theta}}, x, \theta > 0$

We want to est. θ .

$T(\underline{X}) = \sum_{i=1}^n X_i = \bar{X}$ is an unb. est., since

$$E(X_i) = \theta \quad \forall i = 1, \dots, n$$

$$\checkmark \text{ CRLB : } \text{Var}(T(\underline{X})) \geq \frac{\theta^2}{n}$$

$$\text{But we know, } \text{Var}(T(\underline{X})) = \frac{\theta^2}{n}$$

$\therefore T(\underline{X})$ is the unique UMVUE.

Q) In example 3, is $T(X) = X$ the unique unb. est. for θ ? ($n=1$ in this case)

A) Let $T(X)$ be an unb. est. for $\theta \therefore E(T(X)) = \theta$
 $\Leftrightarrow \int_0^{\infty} T(x) e^{-\frac{x}{\theta}} dx = \theta^2 \quad \forall \theta > 0$

(Doesn't it look like a Laplace transform?)

For reasons we wouldn't discuss now, we can say:
 $\int_0^{\infty} T(x) e^{-x/\theta} dx = \theta^2 \quad \forall \theta > 0 \iff T(x) = x.$

$\therefore T(x) = x$ is the unique unb. est.

* Sufficient Statistic: Let $X_1, \dots, X_n \sim P_{\theta}$, $\theta \in \Theta$
 $\underline{X} = (X_1, \dots, X_n)$. $T(\underline{X})$ is some statistic.

$T(\underline{X})$ is called a sufficient statistic if the conditional distribution of $(X_1, \dots, X_n)$ given T is free of θ .

Ex4: $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Bern}(\theta)$, $\theta \in (0, 1)$, $\underline{X} = (X_1, \dots, X_n)$

$T(\underline{X}) = \sum_{i=1}^n X_i$ is sufficient, because:

$$P\{X_1 = x_1, \dots, X_n = x_n \mid T = t\} = \begin{cases} \frac{1}{\binom{n}{t}} & \text{if } t = \sum_{i=1}^n x_i \\ 0 & \text{o.w.} \end{cases} \quad (\# \text{ How?})$$

Ex5: $\underline{X} = (X_1, \dots, X_n)$, $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Poi}(\theta)$

$$T(\underline{X}) = \sum_{i=1}^n X_i$$

$$P\{X_1 = x_1, \dots, X_n = x_n \mid T = t\} = \begin{cases} \frac{t!}{x_1! x_2! \dots x_n!} \left(\frac{1}{n}\right)^t & \text{if } t = \sum_{i=1}^n x_i \\ 0 & \text{o.w.} \end{cases} \quad (\# \text{ How?})$$

Ex6: $X_1, \dots, X_n \stackrel{iid}{\sim} \frac{1}{\theta} e^{-x/\theta}, x, \theta > 0$

$$\underline{X} = (X_1, \dots, X_n)$$

$$T(\underline{X}) = \sum_{i=1}^n X_i$$

Show that, T is sufficient.

(Hint: 1st do for $n=2$, the conditional distⁿ will turn out to be $\propto$)

* Let $X_1, \dots, X_n \stackrel{iid}{\sim} P_\theta, \underline{X} = (X_1, \dots, X_n), \theta \in \mathcal{H}$

Let $T(\underline{X})$ be suff. The $T(\underline{X})$ is called complete if $E_\theta[g(T)] = 0 \forall \theta \Rightarrow g(T) \equiv 0$

- T is suff. & complete. If $h_1(T)$ & $h_2(T)$ are 2 unbi. est. we must have $h_1(T) = h_2(T)$.

Why? : $E[h_2(T) - h_1(T)] = 0 \forall \theta$

$$\Rightarrow (h_2 - h_1)(T) \equiv 0$$

$$\Rightarrow h_2(T) = h_1(T)$$

Q) Are the suff. ~~est~~ ^{statistics} of Ex4 & Ex5 complete?

A) # Check it yourself using polynomial argument.

Note: In both Ex4 & Ex5, the suff. stat. is complete.

Q) What about completeness in case of Ex6.?

A) We will end up w/ yet another Laplace transform, whose prop. we have to accept as a result.